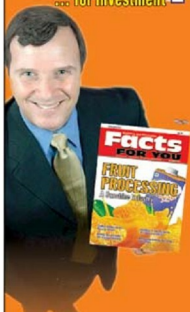


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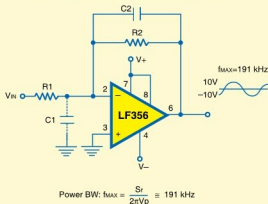
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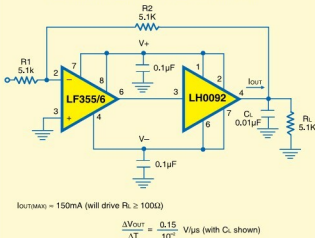
## Typical Applications (Continued)

### Wide BW Low Noise, Low Drift Amplifier



Parasitic input capacitance  $C_1 = (3\text{pF for LF155, LF156 and LF357 plus any additional layout capacitance})$  interacts with feedback elements and creates undesirable high frequency pole. To compensate add  $C_2$  such that  $R_2 C_2 = R_1 C_1$

### Boosting the LF156 with a Current Amplifier



No additional phase shift added by the current amplifier

Datasheets often give application circuits and other relevant information. Also, detailed application relevant information is available in the Application Notes supplied by the manufacturer

Fig. 8: Some application circuits of op-amp type LF156/256/356

some cases, spread in values could be as much as an order of magnitude. To substantiate the statement, the datasheet of C122-series of SCRs from RCA tells that its  $dv/dt$  rating could be anywhere between  $10\text{V/s}$  and  $100\text{V/s}$ . Similarly, the gate-source threshold voltage of power MOSFET type IRF250 could vary from  $2\text{V}$  to  $4\text{V}$  (Fig. 7).

**Application circuits.** Application circuits are usually included only in the detailed version of the datasheet.

These are always an asset to the designers, more so for those who are a little short of design experience. Quite often, expressions used to calculate different component values in a given application circuit are also presented.

Fig. 8 shows some of the application circuits for operational amplifier LF356 given in its datasheet. The information contained in Fig. 8 is reproduced from the detailed datasheet of LF 356/357 series of FET input op-amps.

## To sum up...

This part of the article highlighted the importance of reading a datasheet and the host of technical information contained therein. In particular, it focused on salient features that are common

to the datasheets of all categories of devices illustrating the significance of different parameters with practical examples. The next part of the article will focus on semiconductor diodes.

...To be continued

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